

Wenlin Yi

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EDUCATION

MEng in Electrical and Electronic Engineering

Sept. 2023

Imperial College London, 2:1 Honours

London, United Kingdom

- Relevant modules: Computer Architecture, Digital System Design, Communication Network, Self-Organising Multi-Agent Systems, Adaptive Signal Processing with Machine Intelligence, Deep Learning, Hardware and Software Verification, Embedded Systems, Integrated Analogue Circuit, Optimisation, Control Theory, Mathematics, Large Dimensional Data Processing.

WORK AND RESEARCH EXPERIENCE

Intern, Fundamental Research and Platform Development Center, *CRRC Zhuzhou Locomotive Co., Ltd.*

Feb. 2026 – Apr. 2026

Zhuzhou, China

- Conducted technical investigations on TSN (Time-Sensitive Networking) protocols across multiple demanding industry applications, including satellite communications and computing power network. Improving performance and functional safety in Ethernet-based local area networks (LANs), integrating control function hardware IPs into customised RISC-V architecture.

Research Engineer, Computational Theory Team, CPU Group, *Huawei Technologies Research & Development (UK) Ltd*

Jun. 2024 – Dec. 2025

Cambridge, United Kingdom

- Developed lock-free, flow-balanced, faulty node tolerant Network-on-Chip routing protocols and evaluation methodologies. Integrated to ESL model on C++ for testing and performance evaluation.
- Developed an SAIL-like automated tool to translate ARMv8, v9 ISA to System Verilog. The output of the tool maximally resembles the input ASL coding style while maintaining human-readability and full functional correctness. It is deployed as reference to run coverage oriented tests and facilitate large-scale verification methodology development including innovative state space compression algorithm design;
- Optimised a proprietary automatic theorem prover to cut verification cycle on PEs from years to months; Formally verified floating-point PEs for Huawei AI-Cores, Awarded the Star of Cambridge prize;
- Supported the optimisation of Mat-Mul microarchitecture performance, contributing to algorithmic exploration (matrix tiling, scheduling, hardware-aware mapping) and performance profiling. The joint effort yielded a speed-up of $\geq 20\%$ on 80% of test cases.
- Supported the design and implementation of a render tree caching strategy, with contributions to memory accessing algorithms and memory behavior profiling. Achieved $\sim 80\%$ reduction in data loading time and significant IPC gains.

Discovering Neural Responses to Continuous Speech **Final Year Project, Imperial College London**

Nov. 2022 – Aug. 2023

London, United Kingdom

- Applied data-driven time-frequency methods including multivariate ridge regression, discovered the linear temporal response pattern of EEG to speech signals with 93% accuracy on 16 candidates combined;
- Used reinforcement learning in a refinement loop to alpha search the 13 relevant channels out of 62, extracted the feature-rich mode of these signals, attempted feature-orthogonal mode separation, precisely reconstructed and localised the responsible brain region with high definition, cutting computing time from 2~3 hours to 1~5 minutes per candidate's data;
- Collected and processed the time sequence signals and built machine learning based models to recover the attention and speech envelope(vowels) from EEG signals and predict EEG signals from speech;

'Escape the Pitt' Game and Intelligent Player Agent Oct. 2022 – Dec. 2022
Coursework submitted to Self-Organising Multi-Agent Systems,
Imperial College London London, United Kingdom

- Within the collaborative survival game environment “Escape the Pitt,” the simulation explores how heterogeneous intelligent player agents achieve cooperation and resource allocation through communication, social network formation, collective decision-making, and leadership election under resource scarcity.
- Designed and implemented the game infrastructure; Developed and realised an agent strategy that maximizes collective goal completion and collective intelligence; Constructed dynamic social network trust model to quantify collaborative credibility of other agents; Developed an adaptive combat decision algorithm based on Q-learning; Established a leadership strategy targeting social welfare maximisation. This agents’ algorithm effectively enhanced group survival rates.
- This simulation framework and analytical methods can be directly applied to evaluate multi-stakeholder technology transaction ecosystems, asset pool management, and the dynamic evolution of exchange networks. Providing agent-based modelling (ABM) methodological support for risk assessment and collaborative benefit analysis in valuation.

FPGA Accelerated Machine Learning Research Projects Jan. 2022 – Sept. 2022
Mapping CNNs to FPGAs, Summer Research,
Imperial College London London, United Kingdom

- Researched and benchmarked convolutional neural networks and explore edge conditions of the existing ConvNet to FPGA mapping framework on simulation. Documented and improved user experience.

FPGA Accelerate Super-resolution Network,
Imperial College London London, United Kingdom
Arm Cambridge, United Kingdom

- Found severe bottlenecks of ESPCN super-res net and built hardware IPs to map the parameterised bicubic interpolation layer onto FPGA to enable real-time video enhancement, the tool flow accommodates any size of video and enhances quality up to 3 folds depending on hardware resources.
- Made quantisation benchmarks from both TensorFlow and PyTorch, increase inference speed by 21% with less than 1% peak signal-to-noise ratio drop.

Software Intern, Jetstack Ltd London, United Kingdom, Jun. 2021 – Sept. 2021

- Refined and updated the use of protocol tokens for the certification manager and agents. Replaced existing infrastructure on the echo server and customer server, and improve the user’s onboarding experience.

Undergraduate Teaching Assistant, Imperial College London Jan. 2021 – Mar. 2021

- Assisted the department’s Mathematics and Waves courses in core teaching roles including seminars and class discussions, provided presentation and problem sheets reviews.

SKILLS AND INTEREST

- Technical: Python, C++, C, SystemVerilog, Golang, Rust, Haskell; Linux; SVA, Dafny, Isabelle, Coq; UVM, JasperGold; Verdi, Vivado, Vitis, Verilator, Quartus Prime, ModelSim; Bash, Make; MATLAB.
- Language: English – Proficient, IELTS 7.0 (2017), Chinese (Mandarin) – Native, French – Elementary.
- Activities: Imperial College Robotics Society, Chinese Debating team, Poker Club, Badminton, Tennis, Table Tennis, Gym.